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Sampling Coherent Weather Fields from Probabilistic Foundation Models

A REPORT PRESENTED

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Abstract

Operational forecasts traditionally express uncertainty by running a physics-based model many times from perturbed starting conditions. The WeatherGenerator branch studied here instead outputs a four-component Gaussian mixture at every grid location, so the forecast itself carries the uncertainty. However, the mixtures are produced independently, and sampling each location separately gives a noisy field unlike any real weather pattern. In this report we develop a joint sampling method that selects one value at every location, true to that location's distribution whilst varying coherently across the grid. Rather than recovering the discarded covariance, it imposes a chosen smoothness assumption to build a plausible field from the mixtures. The method is validated on a synthetically constructed testbed whose mixtures are specified in advance, then tested on real 48-hour forecasts. Compared against several baselines, the sampled fields stay smooth without taking values the mixtures consider unlikely, and nearly every value sits centrally in its own distribution. The comparisons carry uncertainty estimates and hold across 12 start dates spanning the test year's seasons. The sampler is most valuable where mixtures hold several distinct plausible values and a field cannot be read from the most likely values alone.

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Chapter 1

Introduction

1.1 Motivation

Weather forecasts inform decisions across agriculture, energy, transport and emergency planning, and preparing for droughts and heatwaves depends not only on the most likely outcome but on how confident that prediction is. Operational ensemble forecasting expresses this confidence through the spread of an ‘ensemble’, integrating a physics-based model many times with perturbations to its starting conditions and formulation. More recently, machine-learning models have begun to supplement or replace this repeated simulation. Some generate whole ensembles of fields, whilst others attach a probability distribution to the predicted value at each location. The WeatherGenerator branch studied in this work produces distributions of this second kind.

The model’s standard head outputs a single value per location, which does not propagate the uncertainty of its prediction. The head studied here instead outputs a Gaussian mixture of four weighted ‘bell-shaped’ components, which carries that uncertainty through and can hold several distinct plausible values at once. However, due to computational constraints the model independently produces a distribution at each location, meaning the covariance between neighbouring locations is discarded. As a result, fields cannot be trivially sampled because independent draws at each location yield the speckled field of Figure 1.1, which no real weather pattern resembles.

Many of the quantities forecasts exist to provide are, moreover, properties of a whole field rather than of any single location. For example, the total rainfall over a river catchment, the extent of a heatwave or the combined output of solar farms across a region. Each depends on how locations vary together, so such quantities can only be drawn from fields that could actually occur. This work therefore develops a method that assembles the independently produced mixtures into one coherent field, so that the mixture head’s richer output is not lost at the final step.

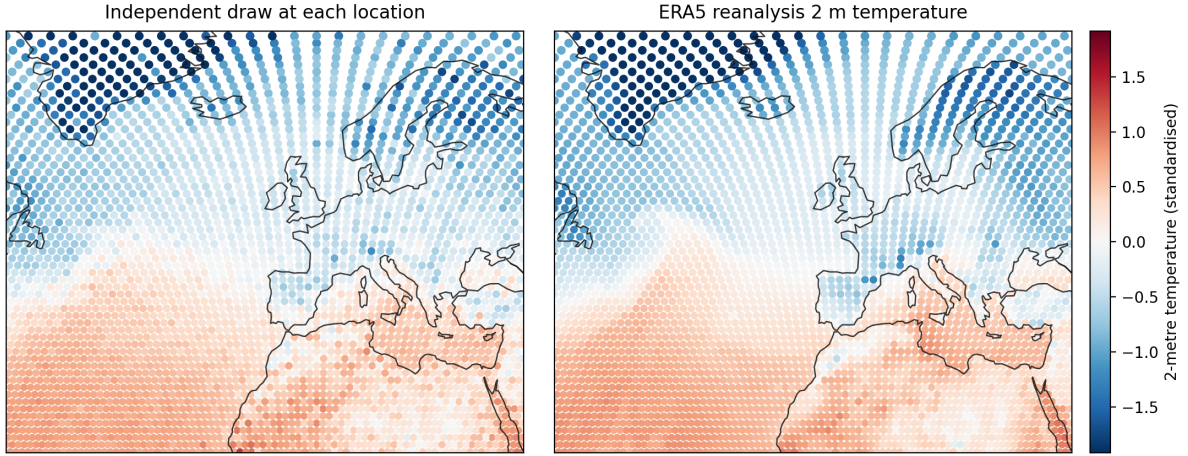


Figure 1.1: A single field drawn independently from each location’s emitted distribution (left) carries fine-scale speckle that no real field exhibits, whereas the corresponding ERA5 reanalysis 2-metre-temperature field [7] on the same O96 grid (right) varies coherently across space. Both are shown over the North Atlantic and Europe on a shared standardised scale; the ERA5 panel is valid at 00:00 UTC on 3 November 2023.

1.2 Aims and approach

The aim of this work is to sample the distributions the model produces so that the resulting field is physically plausible. Plausibility is measured as agreement with each location’s own distribution and with the smoothness assumption that the method deliberately makes.

The difficulty of sampling such plausible fields depends on how spread out the mixtures are and how that spread changes as the forecast reaches further ahead. The forecasts studied reach up to two days out, where a single value often still dominates a location, so the interest lies in measuring how often, and where, genuine mixture alternatives appear in emitted forecasts.

The mixtures are then sampled to create one field that stays true to each location’s own distribution whilst varying coherently in space. Smoothing the field of most-likely values also looks coherent but can wash values into the low-probability gaps between a mixture’s components, so coherence must come without that price.

The method is first exercised on a small synthetic problem whose mixtures are constructed by hand, where the coherent underlying distributions are known in advance and any washing-out or merging is directly visible. It is then tested on real forecasts up to two days ahead, where the mixtures are more spread out than at the earlier forecast times tested. The model can also reconstruct a state it has already seen, an easier case which is used to check the machinery first (Section 5.2).

1.3 Contributions and scope

The main contribution in this report is the joint-sampling method itself. It reaches coherence without a naive smoother’s failure modes of ‘smearing’ values into unlikely gaps or collapsing genuine alternatives, and improves on the smoothed most-likely-field baseline at matched smoothness (Section 5.4).

The smoothness the method imposes is an assumption of ours, not a dependence recovered from the model, which discarded the correlation between locations by design. The sampled fields can be shown to follow this smoothness assumption, but this does not show that the assumption itself is correct.

We also don’t attempt to recreate real weather data, as the model’s mixtures are the only ground truth available, and realism against the observed atmosphere is left to future work. The distributions themselves also partially motivate the problem. They rest almost entirely on a single component when the model reconstructs a state it has seen, and soften across more components as the forecast reaches further ahead, making the two-day forecast a more difficult sampling case.

Chapter 2

Background

2.1 Probabilistic weather prediction and foundation models

The model studied in this report is the WeatherGenerator, a foundation weather model developed at ECMWF [14]. AtmoRep [10] and Aurora [1] are recent related foundation models. Such models often output a single best value per location, when trained by minimising mean squared error, this estimate is drawn towards the conditional mean. GenCast [11] instead generates whole ensembles of fields through a denoising ‘diffusion’ process, whereas the WeatherGenerator branch studied here attaches a marginal distribution to each location individually. The model is trained on the ERA5 reanalysis dataset [7], a decades-long gridded reconstruction of the atmosphere that combines observations with a numerical weather model. For this work, ERA5 is provided on an O96 reduced Gaussian grid, which supplies the locations on which the real forecast fields live.

In WeatherGenerator, the encoder assimilates the input data into latent summaries on a fixed global grid of equal-area cells, a forecasting engine advances that latent state forward in successive six-hour steps, and a decoder reads out each output location from the latent summaries of that location’s own cell and its immediate neighbours. The branch studied here replaces the default single-value read-out with a head outputting a whole probability distribution per location. Because jointly decoding all forty thousand locations with their correlations is too computationally expensive, each location’s distribution is produced on its own, leaving no correlation between the outputted locations.

The sampler produced in this report takes those per-location distributions as fixed input rather than producing them. It is not itself used as a forecasting model and so we don’t attempt to characterise forecast quality.

2.2 Gaussian mixture output heads and multimodality

The distribution emitted at each location is a Gaussian mixture model (GMM) of $K=4$ weighted ‘bell-shaped’ components,

$$p_i(x) = \sum_{k=1}^K \pi_{ik} \mathcal{N}(x; \mu_{ik}, \sigma_{ik}). \quad (2.1)$$

Here $p_i(x)$ is the predicted density of the value x at location i ; the k th component carries a weight π_{ik} , a centre μ_{ik} and a spread σ_{ik} , and the weights sum to one.

A mixture is used rather than a single bell curve because, at one place and time, the atmosphere can allow for two or more distinct plausible values, for instance a front that either has or has not passed. More than one possible value of this kind is what is meant by the distribution being multimodal. However, the weights can also concentrate almost entirely on one component, leaving a near single-valued ‘one-hot’ distribution. This collapse is the behaviour observed in the near-deterministic reconstruction regime studied here.

The literature includes a few potential methods for filling in the lack of covariance between locations. Both Ensemble Copula Coupling [13] and the Schaake shuffle [3] join separate distributions through a ‘copula’, which prescribes only how their relative rankings move together, and both reorder the members of an ensemble that must already exist. The problem for our research is that we do not have an ensemble to reorder, so the two methods answer a different question.

Chapter 3

Sampling Methodology

3.1 Problem statement and the consistency requirement

The studied WeatherGenerator branch outputs a Gaussian mixture p_i of $K=4$ weighted components at each of the N grid locations, with no covariance between neighbouring locations. The task is therefore to choose a single value at every location so that the field is spatially smooth whilst each chosen value stays plausible under its own distribution.

This requirement is termed ‘consistency’ with the emitted distributions, meaning each chosen value must remain likely under its own location’s mixture. It is quantified by the mean per-location negative log-likelihood of the chosen field,

$$\text{NLL}(x)/N = \frac{1}{N} \sum_{i=1}^N -\log p_i(x_i), \quad (3.1)$$

where N is the number of grid locations and $p_i(x_i)$ is the chosen value’s density under its own mixture. A lower value indicates a field more consistent with the mixtures. Neighbouring locations must be coupled on the values they take rather than on the mixture component that produced them, as that component index label has no shared meaning across locations.

Many different fields are equally consistent under this measure, so smoothness is used as the parameter that decides which one is produced. This reflects the restricted spatial variation of real weather, since temperature does not usually change by ten degrees over a few tens of kilometres. Real fields do contain sharp boundaries such as weather fronts, which the smoothing tries not to erase.

The regularity this smoothness imposes is therefore a result of the method, and does not represent a between-location correlation recovered from data the model discarded.

3.2 Baselines

The sampler is judged against five reference fields, each exposing a distinct way a single field can fail. None are offered as methods in their own right.

- **Independent draw:** samples each location’s mixture separately. Every value is plausible under its own distribution, but nothing ties neighbouring values together, which produces a noisy ‘salt-and-pepper’ field.
- **Per-cell MAP:** takes each location’s most likely component centre, the MAP, and leaves it unsmoothed. The result is a jagged field near the consistency floor, with no spatial coherence of its own.
- **Smoothed MAP:** smooths that same MAP field by repeatedly averaging over the neighbour graph. It looks coherent but ‘smears’, inventing in-between values the model considered unlikely.
- **Mixture mean:** takes at each location the mean of its whole mixture. The resulting field is smooth, but the mean of a two-peaked distribution falls in the low-probability gap between the peaks, so the field is often implausible exactly where the mixtures are most spread.
- **Smoothest consistent field, a^* :** the smoothest field found whilst every location keeps to one of its own component centres. Locations are switched one at a time to the centre that best matches their neighbours, until no single switch reduces the squared neighbour differences.

The Smoothed MAP field is the critical baseline comparison, as its roughness level is later used set the matched target at which the method is tuned. The comparison setting of the blur uses ten averaging passes at a step of 0.1, which is chosen only to give a visibly coherent reference field, as the aim of the roughness matching is just to make the comparison relative, the set value itself is actually fairly arbitrary.

3.3 Mode-consistent joint sampling

Two samplers are built on the requirement of Section 3.1. One is the primary continuous-value method, the other a mode-based variant run on the synthetic testbed.

Joint MAP. The primary method builds one field by minimising an objective that balances consistency against smoothness,

$$J_\lambda(x) = \text{NLL}(x)/N + \lambda \frac{S(x)}{|E|}. \quad (3.2)$$

The first term is the consistency measure of the previous section, the second a roughness penalty S that grows as neighbouring values differ, divided by the number of neighbouring-location pairs $|E|$. The graph defining those pairs, and S itself, are set out in Section 3.4. The weight λ sets the balance between the two terms, so a field with a low J_λ is both consistent and smooth.

The objective is minimised using gradient descent. This is done using the closed form gradient from the mixture responsibilities r_{ik} which are defined as the share of the mixture density that each component takes responsibility for. Descent runs with the Adam optimiser [9] from a ‘warm start’ at each location’s most likely value, over a few random restarts with the lowest- J_λ field retained. The restart count, step budget, learning rate and jitter are given in Appendix A, and the gradient formula and comparisons with certified optima are in Appendix D. The Smoothed MAP baseline is also constructed by blurring that same most-likely field, so both set out from one starting point and neither gains an advantage from its initialisation.

Mode-selection MRF. A field is ‘mode-consistent’ when every value sits on a peak of its own distribution, and the mode-based variant makes that idea the whole method. It works in two stages, candidate peaks of each location’s mixture are located first, and one peak is then selected per location by minimising

$$J_\beta(a) = \frac{1}{N} \sum_i u_i(a_i) + \beta \frac{1}{|E|} \sum_{(i,j) \in E} (m_{i,a_i} - m_{j,a_j})^2, \quad (3.3)$$

where a_i names the peak selected at location i , $u_i(k) = -\log p_i(m_{ik})$ scores a peak by how likely it is under its own mixture, and the second term penalises neighbouring selections that sit at distant values, with the temperature β setting the strength of this penalty. The candidate peaks m_{ik} themselves are found beforehand by the mean-shift fixed point [2],

$$v \leftarrow \frac{\sum_k r_k(v) \mu_{ik} / \sigma_{ik}^2}{\sum_k r_k(v) / \sigma_{ik}^2}, \quad (3.4)$$

where $r_k(v)$ is the responsibility of component k at the free point v , the same quantity as r_{ik} in (D.1) but evaluated at v rather than at the data value x_i . When iterated from each component’s mean, it climbs to a peak of the mixture.

A sweep across a range of temperatures β is run on the synthetic testbed, where underlying distributions are known. On the real forecasts the variant degenerates, over the forecast times tested the emitted mixtures still place almost all their weight on one peak per location, so re-assigning any location to a lower-weighted peak costs far more likelihood than the neighbour term can offset. Mode-selection MRF is only included in the forecast audit of Chapter 5 as a reference field, not a competitor. The degeneracy is

only a concern when choosing peaks discretely, however Joint MAP enjoys the freedom of continuous placement of each value around those peaks.

Joint MAP is therefore the primary sampling technique, the continuous-value method that scales cheaply to the full grid. Its weight λ and the variant’s temperature β are both fixed for comparative purposes by the selection rule of the next section.

3.4 Smoothness-regularised selection and temperature

The roughness penalty measures how much neighbouring values differ. Each location’s neighbours are its eight nearest, the surrounding cells on the synthetic testbed’s square grid and the closest locations along the sphere’s surface on the forecast grid. The penalty sums the squared value differences across those links, compactly a quadratic form in the graph Laplacian L of this neighbour graph,

$$S(x) = x^\top Lx = \sum_{(i,j) \in E} (x_i - x_j)^2. \quad (3.5)$$

On this graph the Smoothed MAP baseline’s blur reduces to a simple form, each of its averaging passes being the update

$$x \leftarrow x - \eta Lx \quad (3.6)$$

with η the step size, which is gradient descent on S alone (the penalty’s gradient is $2Lx$), the same operator as the method’s penalty but with no consistency term. The Smoothed MAP is therefore what Joint MAP becomes with the consistency term removed. Both methods smooth by the same operation, so matching them at equal roughness compares like with like.

The roughness measure itself is designed to be scale-free. The squared difference is first averaged over the graph’s links, giving the edge mean S_{edge} , and that average is then divided by the variance of the field’s values over all locations,

$$\tilde{R}(x) = \frac{S_{\text{edge}}(x)}{\text{Var}_V(x)}, \quad S_{\text{edge}}(x) = \frac{S(x)}{|E|}. \quad (3.7)$$

The roughness is divided by the field’s own variance to prevent optimisation to a trivial state. Without it a method could look smoother simply by shrinking the whole field toward a flat average; $\tilde{R}$ is instead unchanged if the field is rescaled or flattened.

The selected weight λ^* is the smallest λ whose field is as rough as the Smoothed MAP baseline’s. In practice λ is swept over a grid of values, the roughness $\tilde{R}$ of each swept

Measurement	Property secured	Failure caught
Per-location NLL	values stay true to their own distribution	an over-smoothed field in unlikely gaps
Scale-free $\tilde{R}$	the field is no smoother than asked	a speckled, salt-and-pepper field
Off-mode drift and spectrum	smoothness without inventing unlikely values	smearing into dips, and mode collapse
Central marginal-position coverage	values keep to the central mass of their own distributions	a value forced into its own far tail

Table 3.1: The four-measurement evaluation bundle.

field is read off, and the crossing point is found by interpolating between the two nearest settings (against $\log \lambda$). The variant’s temperature β is chosen in exactly the same way. Fixing a roughness target gives every method a well-defined target to be matched at.

3.5 Evaluation protocol

The sampler is judged on four measurements shown in Table 3.1. Two of them, the per-location NLL and the scale-free $\tilde{R}$, have already been explored. The two that remain are set out here. The most helpful is the off-mode drift, which measures how far a field’s values have been dragged off their own peaks into the low-probability gaps between them. This is the smearing failure a smoother tends to create. It is defined as the likelihood a value gives up against its own best peak,

$$\Delta\text{NLL}_i = -\log p_i(x_i) - \min_k (-\log p_i(m_{ik})). \quad (3.8)$$

The reference is the location’s single best peak, its highest-density mode, so a value resting on a legitimate secondary mode still registers a positive drift. To fix a boundary for when a value stops being plausible under its own mixture, a value is counted ‘off mode’ once it drifts by more than half a standard deviation’s worth of likelihood (0.125 nat, the natural-log unit). The synthetic testbed of Chapter 4 applies the same rule as a distance, counting a value off mode more than 0.5σ from its nearest peak.

The final measurement checks where each chosen value falls within its own distribution. The mixture’s cumulative distribution gives that position directly and is shown using the

probability integral transform (PIT) [5],

$$u_i = F_i(x_i) = \sum_{k=1}^K \pi_{ik} \Phi\left(\frac{x_i - \mu_{ik}}{\sigma_{ik}}\right), \quad (3.9)$$

placing the selected value x_i at a quantile u_i of its own mixture. Forecast verification applies this transform to observed outcomes, where an even spread of quantiles indicates calibration. Here we instead apply it to values selected by the sampler, and so we do not expect or look for an even spread. The central-90% coverage is the fraction of locations with $u_i \in [0.05, 0.95]$, read alongside a central-50% band $[0.25, 0.75]$. The coverage is a plausibility check internal to the selected field, not a statement about the calibration of an ensemble. A value resting in a low-density gap between two modes can still hold a central quantile, so catching that failure remains the off-mode drift’s role.

Three descriptors of the emitted mixtures themselves are defined here for Chapter 5’s use. Each reduces to one number how far a location’s mixture weights concentrate on a single component. The **median max- π** takes the largest of each location’s mixture weights and reports its median over locations; the **one-hot fraction** is the share of locations where that largest weight exceeds 0.9; and the **second-mode mass** is the median second-largest mixture weight.

Finally we note that the objective minimises on the axes of NLL and $\tilde{R}$, so scoring well on them shows only that the optimisation worked, not that the field is a good one. We use the off-mode drift as the real test, as it is a measurement the objective was never optimised on.

Chapter 4

Validation on a Synthetic Testbed

4.1 The synthetic testbed

The WeatherGenerator GMM branch emits mixture distributions, the only basis we can use as a ground truth. To help give some more insight, we also built a controlled testbed with known underlying fields before focussing on real forecasts. In the synthetic problem the mixture parameters such as modes and variance are manually set. The testbed is a single weather-like variable on an 8×8 grid, each location carrying a mixture of $K=4$ Gaussian components. Their centres are drawn from smooth quadratic surfaces, shown in Figure 4.1, that drift slowly between neighbours. Each location receives the four surface values in a shuffled order of its own, so a component’s index carries no meaning shared between locations. Additionally, the weights at each location are drawn independently of every other, and all components share one fixed spread ($\sigma = 1$).

This reproduces the real decoder’s outputs, independent per-location GMMs which are then turned into a scalar field by sampling. No single hidden field is therefore the unique correct answer. Success is instead defined by producing a field that stays true to its mixtures whilst remaining spatially coherent.

Every sampled field is measured by consistency, NLL/N , and by the scale-free roughness $\tilde{R}$. The baselines explored above are shown in Figure 4.2. The Independent draw is the noisiest field, the Per-cell MAP the most consistent yet still jagged, and the Smoothed MAP blur reaches the lowest roughness of all whilst paying a higher likelihood cost than a^* .

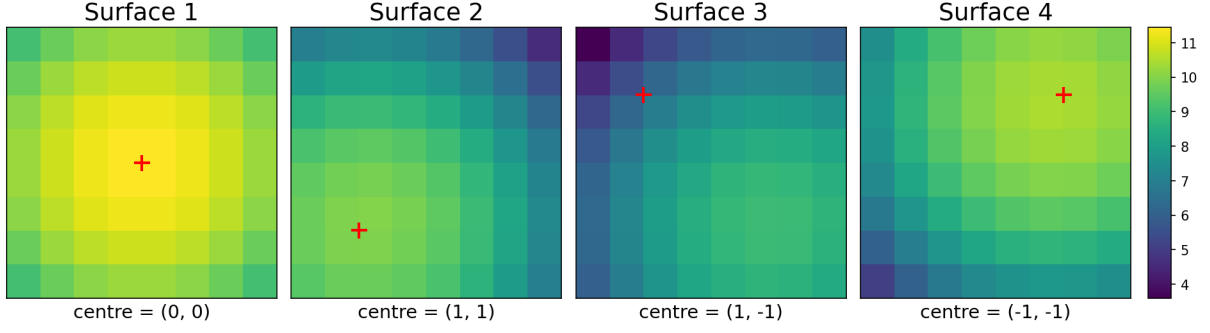


Figure 4.1: The four smooth quadratic surfaces from which each location’s component centres are taken, each marked at its peak.

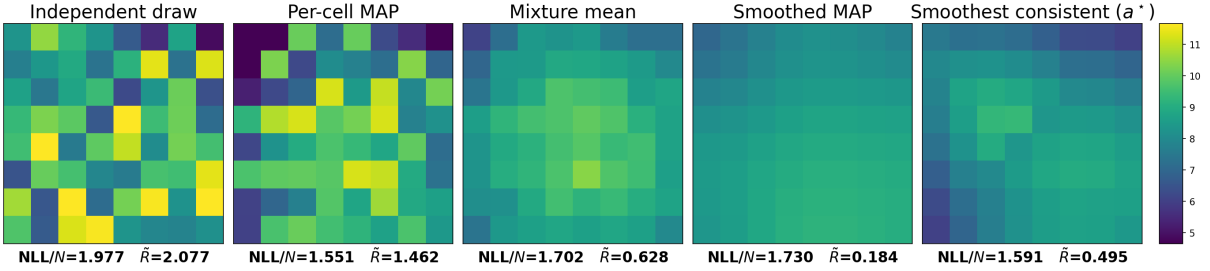


Figure 4.2: Reference fields on the synthetic testbed, each shown with its consistency measure (NLL/N, lower is more consistent) and its scale-free roughness $\tilde{R}$ beneath the panel.

4.2 Method behaviour with controllable modes

Here we explore how far each method can be pushed towards smoothness before values are dragged off their peaks into the unlikely gaps between them. Figure 4.3 tracks that displacement against roughness for the Joint MAP and the Mode-selection MRF, and Figure 4.4 swaps in a different penalty to test whether the smearing that appears is particular to the quadratic form or the price of smoothness itself. The peaks here are manually set, creating a specific regime with which to test the samplers.

Figure 4.3 counts a location as smeared once it is displaced by more than 0.5σ from its nearest peak. Each point carries an interval built from the field itself, the field’s cells are resampled 2000 times and the central 95% range of the recomputed fraction is shown, this is the spread of the measurement within that one field. At matched or lower roughness the Mode-selection MRF leaves fewer locations stranded in the gaps between peaks than the Joint MAP, holding only 6.2% of cells past 0.5σ at the smoothest setting it reaches without smearing, far fewer than the Smoothed MAP blur. The over-smoothed references score high on this off-mode axis, contrastingly, the Per-cell MAP shows far higher roughness, yet offers only a small faithfulness gain over the other baselines ($\tilde{R}=1.462$, near the Independent draw’s 2.077), a weakness of Per-cell MAP in regimes with more equally weighted mixtures. The independently drawn weights make the most likely component

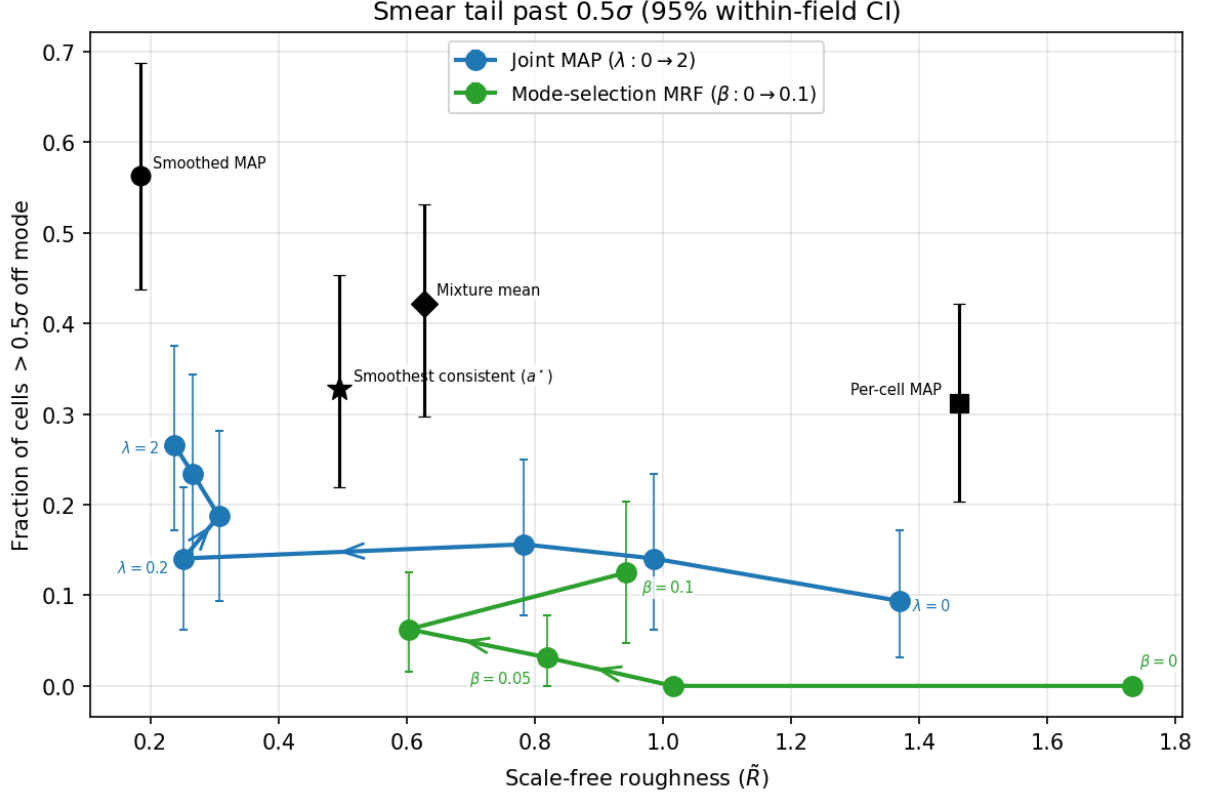


Figure 4.3: Non-smearing on the synthetic testbed: the fraction of cells displaced more than 0.5σ off their nearest emitted mode against scale-free roughness $\tilde{R}$, for Joint MAP on the discrete $\lambda \in \{0, 0.05, 0.1, 0.2, 0.5, 1, 2\}$ sweep and Mode-selection MRF on the compact $\beta \in \{0, 0.025, 0.05, 0.075, 0.1\}$ grid, with the Mixture mean, Smoothed MAP, Per-cell MAP and a^* reference fields; within-field 95% confidence intervals over 2000 cell resamples are shown (Independent draw off-scale; Figure 4.2).

change sharply from cell to cell leading to a rough ‘salt-and-pepper’ sample.

Figure 4.4 is used to answer whether that smearing comes from the quadratic penalty in particular or from forcing smoothness at all. A third method is placed on the same consistency–roughness plane, total-variation smoothing [12]. This penalises absolute rather than squared neighbour differences, solved exactly to its global optimum by a min-cut, so any smearing it shows cannot be blamed on the optimiser. The exact total-variation field stays free of smearing only down to a moderate roughness ($\tilde{R}=0.575$), past that point it stops smoothing and instead collapses towards a near-constant field, holding 6.2% of cells off their peaks, essentially the floor the Mode-selection MRF holds (6.2%). Only the quadratic Joint MAP pushes to a much lower roughness ($\tilde{R}=0.252$) as seen in Figure 4.5. However it pays for it by smearing, leaving 14.1% of cells off their peaks. The trade-off is therefore structural, a property of the penalties themselves rather than of how well the optimiser was run, meaning below a moderate roughness the quadratic penalty smears and the total-variation penalty collapses.

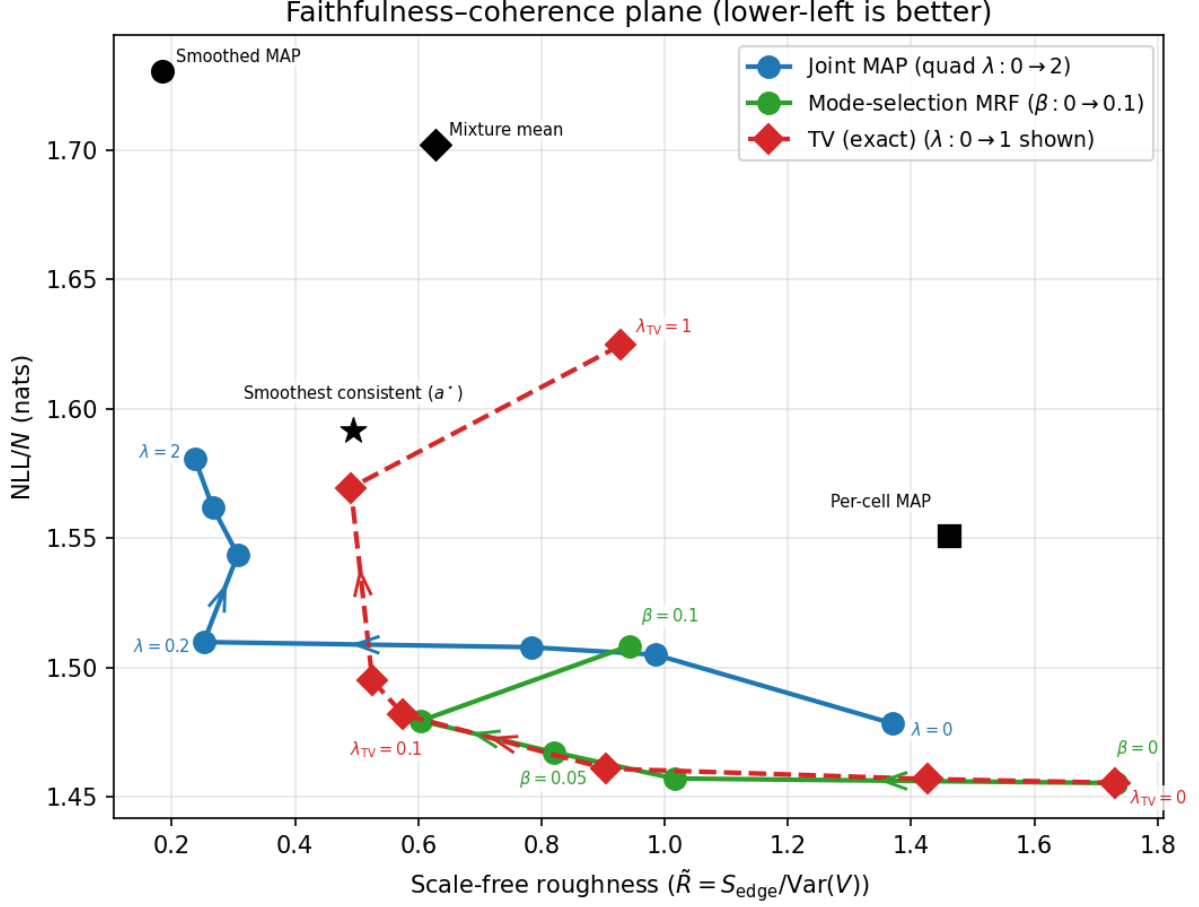


Figure 4.4: The consistency–coherence plane with the exact total-variation frontier (synthetic testbed): NLL/N versus scale-free roughness $\tilde{R}$ for Joint MAP and Mode-selection MRF, sweeps and omissions as in Figure 4.3, and the exact min-cut total-variation frontier ($\lambda_{TV} \in \{0, 0.02, 0.05, 0.1, 0.2, 0.5, 1\}$ shown; collapsed $\lambda_{TV} \geq 2$ excluded), with the Smoothed MAP, Per-cell MAP, Mixture mean and a^* anchors.

4.3 What the testbed establishes, and its limits

The two penalty methods can be tuned to a chosen level of smoothness whilst keeping locations on their peaks, avoiding the likelihood cost the Smoothed MAP blur must pay to reach the same low roughness. However, the testbed is deliberately simple, an 8×8 flat Cartesian grid, with constant-variance GMMs, containing none of the structure of real weather such as fronts. It therefore gives no insight into behaviour on the real spherical grid, or into how well each method samples a real forecast.

Its main role is a check that the methods behave as intended in a controlled regime. However, the independently drawn mixture weighting also gives insight into the sampler’s efficacy in a regime without a single dominant component, found in the 48-hour forecasts of the next chapter.

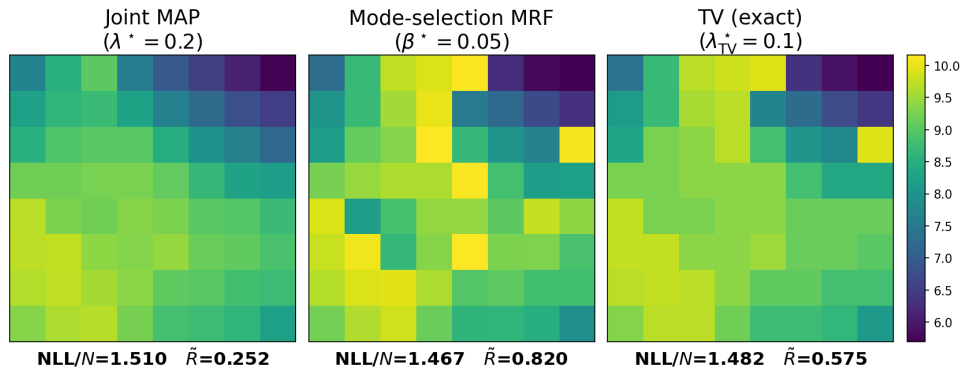


Figure 4.5: All three sampling methods used in Figure 4.4 applied to the synthetic testbed: Joint MAP, Mode-selection MRF and exact total-variation fields at $\lambda^* = 0.2$, $\beta^* = 0.05$ and $\lambda_{TV}^* = 0.1$, respectively.

Chapter 5

Sampling Real Forecast Fields

5.1 Forecast configuration and evaluation protocol

The same consistency-versus-roughness reading and both penalty methods are applied to real 48-hour forecast fields on the full spherical grid. The emitted distributions are from the WeatherGenerator trained end-to-end as a single network on the ERA5 data over its O96 reduced Gaussian grid of $N = 40,320$ locations, six-hourly 2-metre temperature, standardised per channel to zero mean and unit variance using the training archive’s precomputed statistics, with training spanning 1979–2022 so that all of 2023 remains unseen. Every field in this chapter is generated from a 2023 initialisation. Two models were trained in sequence, a masking autoencoder from scratch and a +48-hour forecast model warm-started from it (Appendix A). A separate extraction step then runs the trained forecast model in inference once per start date and saves the mixtures it outputs, and these fixed mixtures are the input the sampler consumes (Appendix E).

Three models from this pipeline appear in this chapter: the main 14-epoch forecast model (on which every headline result is computed), its earlier 6-epoch checkpoint (kept as a training-evolution control), and the autoencoder itself, which supplies the near-deterministic reconstruction limit of Section 5.2. All three models are held fixed throughout, so the rest of the chapter tests the behaviour of the sampler on a fixed set of mixtures rather than the fit of a model.

Robustness is assessed by running the 14-epoch model from 12 start dates, the first day of each month of 2023, which places three initialisations in each of the four seasons so the results span the full annual range rather than one favourable case. These runs vary the weather situation the model forecasts from whilst the model itself is unchanged, so they measure how stable the sampler’s behaviour is across situations rather than how well the model generalises. Because every initialisation falls within a single year, the twelve runs cover behaviour across that year’s seasons only. Generality across multiple years, or against long-term averages, is not tested.

Hours ahead	median max- π		one-hot frac.		2nd-mode mass	
	6 ep	14 ep	6 ep	14 ep	6 ep	14 ep
+6h	0.953	0.944	71.3%	67.9%	0.042	0.044
+48h	0.803	0.766	11.2%	9.3%	0.151	0.178

Table 5.1: Mixture-weight softening at the two headline steps (+6 h, +48 h), 6-epoch vs 14-epoch, for the base-case initialisation; the full curve across all steps is Figure 5.1. Pinned definitions in Section 3.5.

Two kinds of interval accompany the results that follow. The ‘within-field’ interval of Chapter 4 reports how much a quantity varies from place to place inside one field, not any uncertainty across forecasts. The second kind, an across-initialisation range, is the smallest-to-largest spread of a quantity over the 12 initialisations, and reports how much the trained model’s behaviour moves as the weather situation changes. One initialisation, 1 November 2023, on which the evaluation was developed, is kept as the base case. Each of the figures and tabulated point values is based on it, with across-initialisation ranges quoted alongside.

5.2 The emitted distributions: from reconstruction to two days out

As the forecast reaches further ahead the emitted mixtures soften steadily. For the 14-epoch model the median largest mixture weight (median max- π) falls from 0.944 at +6 h to 0.766 at +48 h, with the near-single-valued (‘one-hot’) fraction falling and the typical second-mode weight rising in step, shown in Table 5.1. Placing the 6-epoch checkpoint alongside the 14-epoch model (Figure 5.1) shows that the more-trained model softens further with its +48 h median max- π sitting below the 6-epoch value of 0.803. Hence the spread comes from forecasting further ahead rather than from an unfinished model.

Throughout this chapter a cell is counted ‘bimodal’ when some pair of its components each carry a mixture weight of at least 0.1 and their centres sit more than 2 times the larger of their two spreads apart, the separation at which the density itself develops two peaks with a genuine low-probability gap between them. Cells of this kind stay in the minority (6.3%) even at +48 h, and most are at tropical latitudes, within 23° of the equator, which is consistent with literature that suggests forecast ability is often lower for tropical weather, especially near convectively active latitudes [4, 6].

The easy end of this range is the autoencoder’s reconstruction limit, where the model completes a state it is partly shown, so the mixtures are almost single-valued and the predicted field is essentially known. The sampler behaves as intended here, the roughness-

Forecast Softening: 6-Epoch vs 14-Epoch

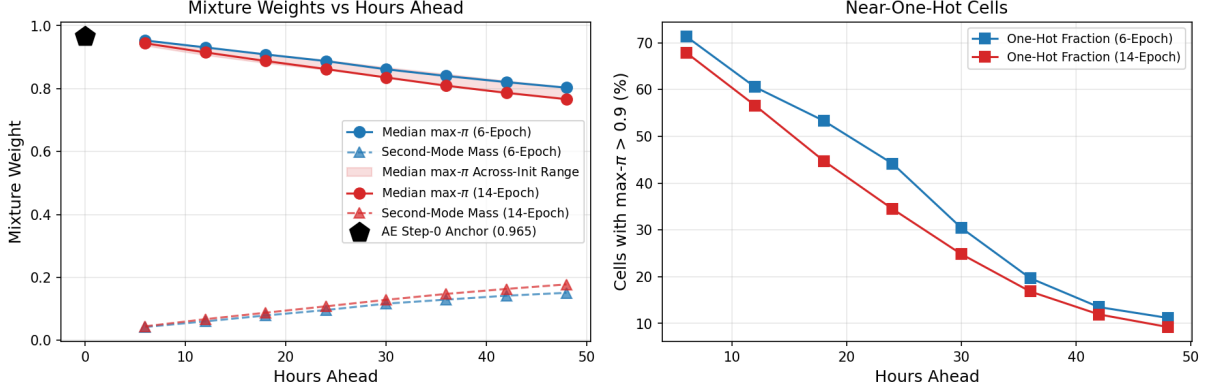


Figure 5.1: Mixture-weight softening as the forecast reaches further ahead, 14-epoch model (red) versus the 6-epoch checkpoint (blue). *Left*: median max- π (circles) and median second-mode mass (triangles), with the autoencoder step-0 reconstruction value as the black pentagon at zero hours ahead. The shaded band is the across-initialisation min-max range of the *median max- π* only (12 first-of-month 2023 cases); the second-mode and one-hot curves are base-case point estimates. *Right*: one-hot fraction ($\max \pi > 0.9$).

matched weight selected cleanly at $\lambda^* = 93.7$, and at that matched roughness Joint MAP leaves fewer cells off their best emitted mode than the Smoothed MAP blur (16.9% against 46.6%), a gap whose 95% within-field interval excludes zero.

5.3 Selecting the smoothness weight at each forecast step

The forecast model steps forward in six-hour intervals, and the sampler is applied to the mixtures emitted at each step, from +6 h out to +48 h. The smoothness weight is re-selected at every step, so that the sampled field’s roughness matches that step’s own Smoothed MAP target, following the matched-roughness approach of Section 3.4.

The selected weights for both models are shown in Table 5.2. At every step the matched weight fell inside the range swept, and the full sweep grids are given in Appendix B. At +48 h the 14-epoch weight is $\lambda^* = 81.3$. It spans only [61.9, 83.0] (mean 72.8) across the 12 initialisations, and reseeding the optimiser’s random restarts moves it by less than 1.6×10^{-3} on the base case.

The autoencoder reconstruction weight ($\lambda^* = 93.7$) is of the same order as the forecast weights, yet the forecast weight drifts downward overall as the forecast reaches further ahead, though not monotonically. As the mixtures broaden, moving a value away from its peak costs less likelihood, so a smaller weight is enough to reach the same matched roughness.

Hours ahead	λ^*	
	6 ep	14 ep
+6h	75.8	116.3
+12h	61.3	91.5
+18h	56.3	97.6
+24h	56.2	97.0
+30h	51.4	88.1
+36h	47.0	78.5
+42h	48.1	80.6
+48h	52.0	81.3

Table 5.2: Selected roughness-matched λ^* (re-selected against each step’s Smoothed MAP $\tilde{R}$ target) for both models; every sweep range contained its matched λ^* . The 14-epoch +48 h λ^* spans [61.9, 83.0] (mean 72.8) across the 12 first-of-month 2023 initialisations; the tabulated 14-epoch value is the base case.

5.4 Joint sampling results at +48 h

For +48 h forecasts the outputted mixtures start to become genuinely varied, meaning there are more grid points with two or more mixture components with significant weight. This step is therefore where we attempt to fulfil the main aim of this work, drawing one field coherently from the independent mixtures without its values being washed into the low-probability gaps between peaks.

Figure 5.2 shows the five different fields produced from one set of mixtures output by the model, together with the ERA5 reference, over a North Atlantic and Europe window. The Independent draw is visibly the noisiest of them, whilst the Per-cell MAP baseline carries the fine-scale texture closest to ERA5’s. Joint MAP at λ^* and the Smoothed MAP blur are deliberately smoother than both, as the blur smooths the Per-cell MAP field directly and λ^* is chosen so that Joint MAP matches its roughness. Compared with ERA5 ($\tilde{R} = 0.0215$), the Per-cell MAP is smoother ($\tilde{R} = 0.0101$) and the Independent draw rougher ($\tilde{R} = 0.047$). This is expected for 48-hour predictions, as forecast uncertainty leads training to favour predictions that leave neighbouring cells smooth rather than sharply varying. The Joint MAP panel at $\lambda=19.0$ is the ‘budget’ setting, the largest smoothness weight that keeps 95% of cells within 0.5σ of their best peak (4.6% off mode, $\tilde{R} = 0.0048$), and it visibly retains more of the emitted texture than the λ^* panel.

However, these panels alone cannot judge the sampler. Even at +48 h, at most locations one component still carries a large share of the total weight, so the Per-cell MAP baseline is expected to perform well in this regime. The ERA5 panel, in turn, shows what actually occurred rather than a target the sampled fields can be judged against, as every other panel is drawn from the model’s predicted mixtures and so is limited by the model’s predictions themselves.

We then return to the matched comparison which is made in the left panel of Figure 5.3, this shows each field’s likelihood cost against its roughness. At the Smoothed MAP roughness target Joint MAP reaches a lower per-location negative log-likelihood than the blur. However, the improvement gap is modest, and these are the axes the method was optimised on, so they cannot settle the comparison alone.

The deciding measure is the smear tail (Figure 5.3, right), the fraction of cells pushed more than 0.5σ off their best emitted mode. On that measure the two methods separate cleanly exactly where a genuine choice between peaks exists. Within the bimodal cells, Joint MAP leaves 9.0% of cells off mode against the blur’s 19.3%, an ordering that holds across all 12 initialisations. These far-displaced cells are where the method starts to falter. The right-hand panel of Figure 5.4 maps where they sit on the globe. The left panel shows how strongly each method’s off-mode cells concentrate on the bimodal subset. Notably, this ratio isn’t an indicator of how many cells are off mode in total, a value on which Joint MAP stays well below the Smoothed MAP (Figure 5.3, right). Within the well-separated ($> 2\sigma$) bimodal cells the Joint MAP ratio sits below one (0.69), rising only modestly under the looser $> 1\sigma$ definition (1.23), meaning the sampler stays mode-respecting when the modes are well separated. When a value is instead counted off mode only when displaced by a full standard deviation, rather than the half used in the figure, the picture inverts, the Joint MAP ratio rising to 1.34 against the Smoothed MAP’s 0.44, so the rare deep excursions do concentrate on the bimodal cells, matching how the quadratic penalty behaved on the testbed of Chapter 4.

As a qualitative check on spatial coherence, the native spherical angular power spectrum is used. Figure 5.5 gives the power the field holds at each angular scale ℓ ,

$$C_\ell = \frac{|a_{\ell 0}|^2 + 2 \sum_{m \geq 1} |a_{\ell m}|^2}{2\ell + 1}, \quad (5.1)$$

where the $a_{\ell m}$ are the field’s spherical-harmonic coefficients and larger ℓ corresponds to finer scales. Every field sits well below the Independent draw noise floor at fine scales, so none of them are dominated by sampling noise, and Joint MAP at λ^* sits just above the Smoothed MAP blur it is matched to, keeping slightly more fine detail. The ERA5 reference agrees with every field at the largest scales but holds roughly 173 times the λ^* field’s power at finer scales. Each spectrum is normalised to unit power over the resolved band ($\ell \leq 191$), so this ratio compares the share of each field’s variance held at fine scales rather than absolute amplitudes.

Finally, from Figure 5.3 we note that the improvement margin of Joint MAP over the Smoothed MAP is stable. At the matched roughness target the blur leaves 32.7% of cells off mode against the sampler’s 13.2%, roughly two and a half times as many, and the same factor holds at the gentler and harsher blur strengths of the left panel, each against Joint

Hours ahead	marg.-pos. 90%	
	6 ep	14 ep
+6h	99.4%	98.3%
+12h	99.6%	98.7%
+18h	99.7%	98.7%
+24h	99.7%	98.6%
+30h	99.7%	98.6%
+36h	99.8%	98.8%
+42h	99.8%	99.0%
+48h	99.8%	99.0%

Table 5.3: Joint MAP central marginal-position coverage at λ^* , at each forecast step and for both models: the share of cells whose selected value lies within the central 90% of its own emitted mixture’s mass. The tabulated value at each step is from the single reference initialisation; the across-init mean and range are quoted in the text.

MAP at approximately matched roughness (21.2% against 8.4%; 46.5% against 18.6%), so the improvement margin is largely independent of the λ weighting applied to Joint MAP. Nor does it depend on the starting state. Across the 12 initialisations, Joint MAP keeps a lower off-mode fraction and a lower likelihood cost than the blur in every case, and the +6 h margins are larger still. At matched smoothness the sampler therefore keeps far more values on their own emitted peaks than the blur.

5.5 Consistency with the emitted distributions

We use the central-90% coverage defined in Section 3.5, as it catches the one failure the earlier measurements could not show, a value forced into the far tail of its own distribution by the smoothing. At +48 h nearly every value Joint MAP selects at λ^* lies within that band (99.1%, across-init range [98.9%, 99.3%]). Along the smoothness sweep in the left panel of Figure 5.6 this coverage holds essentially unchanged through to λ^* , so selecting for smoothness leaves each value’s position within its own distribution intact, with only a slightly increasing likelihood cost (right panel). These values at each forecast step are shown in Table 5.3. Together with the matched-smoothness result of Section 5.4, this shows the Joint MAP-sampled field is both true to the emitted distributions and smooth across space at the 48-hour forecast step.

Forecast +48 h: 2-metre temperature, North Atlantic / Europe

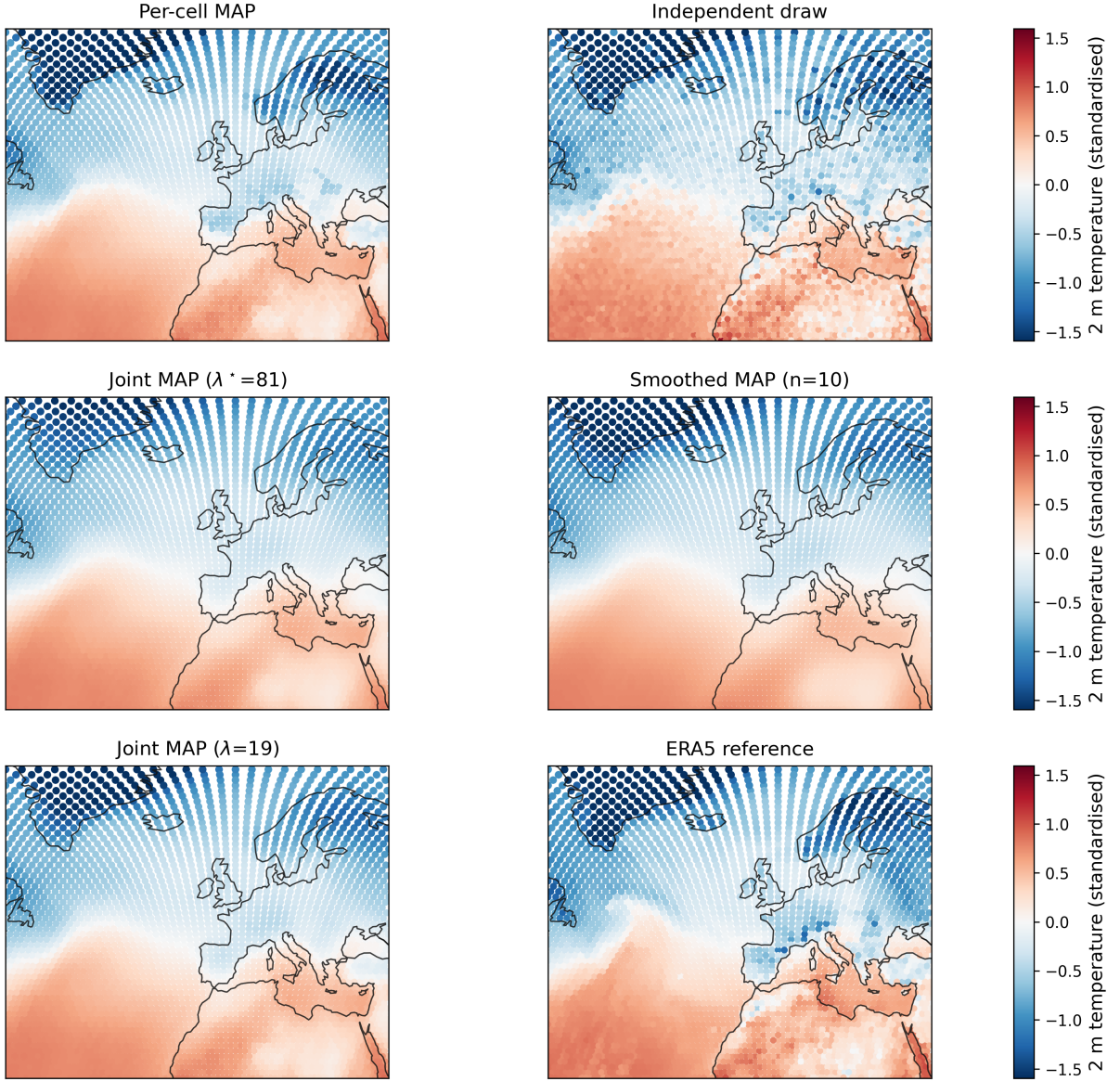


Figure 5.2: Sampled +48 h fields and ERA5 reference for one unseen 2023 snapshot over the North Atlantic / Europe window (14-epoch model, standardised units). Row-wise: the Per-cell MAP baseline and an Independent draw from the emitted marginals; Joint MAP at λ^* and the Smoothed MAP blur ($n=10$) at matched $\bar{R}$; and Joint MAP at the budget setting $\lambda=19.0$ and ERA5. Appendix C carries the whole-globe view and extends it to the reconstruction limit and the +6 h and +24 h steps.

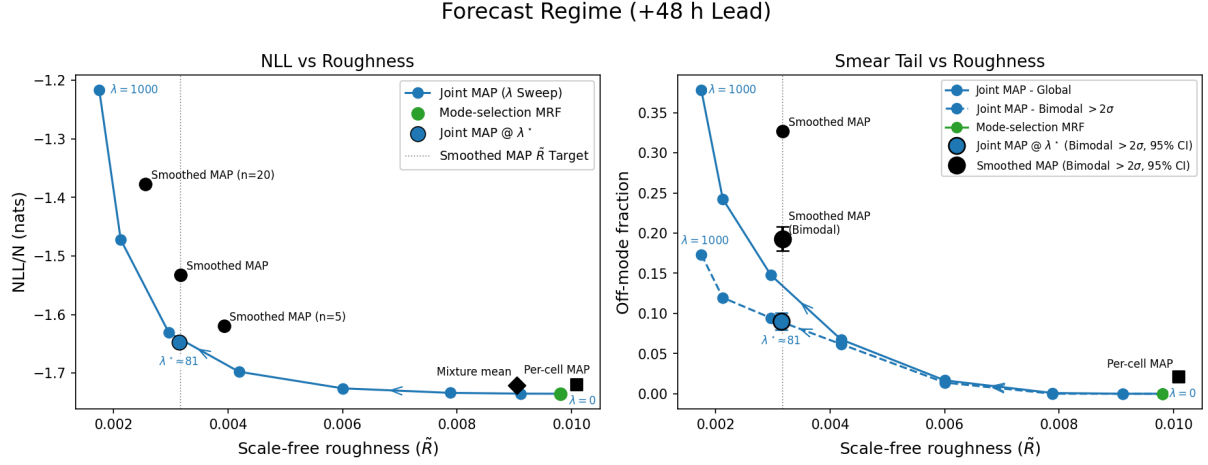


Figure 5.3: The matched- $\tilde{R}$ smear-tail diagnostic, 14-epoch model at +48 h. *Left*: the consistency-coherence plane (NLL/ N vs $\tilde{R}$, linear axis): the Joint MAP λ sweep ($\lambda \in \{0, 0.5, 2, 8, 30, 100, 300, 1000\}$), the Smoothed MAP, Per-cell MAP and Mixture mean baselines, and Joint MAP at λ^* (the black-edged circle) at the Smoothed MAP $\tilde{R}$ target (dotted line). *Right*: the fraction of cells left off their best emitted mode ($\Delta\text{NLL} > 0.125$ nat) against $\tilde{R}$, for the global field (solid) and the bimodal cells (dashed): this shows a separation between methods that the left-hand plane cannot show. The Mode-selection MRF β sweep collapses to a single point (at the sweep origin) in both panels, the degeneracy described in Section 3.3.

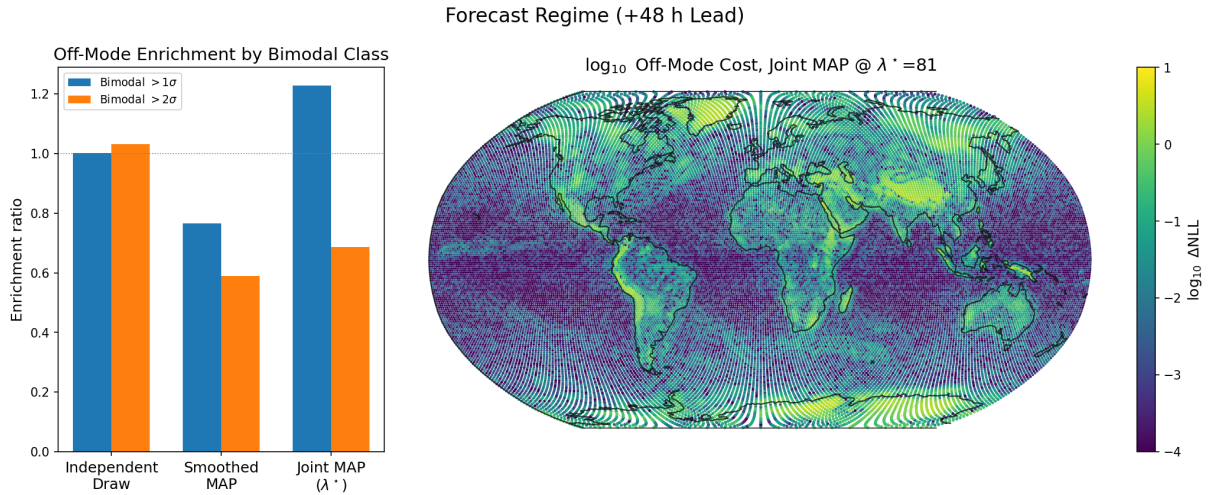


Figure 5.4: Where the smear tail lives, 14-epoch model at +48 h. *Left*: enrichment of off-mode ($\Delta\text{NLL} > 0.125$ nat) cells within the bimodal subsets relative to the global field, for each method. The $> 2\sigma$ bars follow the bimodal definition of Section 5.2 (2,524 cells); the $> 1\sigma$ bars loosen that separation to 1σ (13,994 cells). *Right*: the per-cell $\log_{10} \Delta\text{NLL}$ to best mode for Joint MAP at λ^* , mapped over the globe.

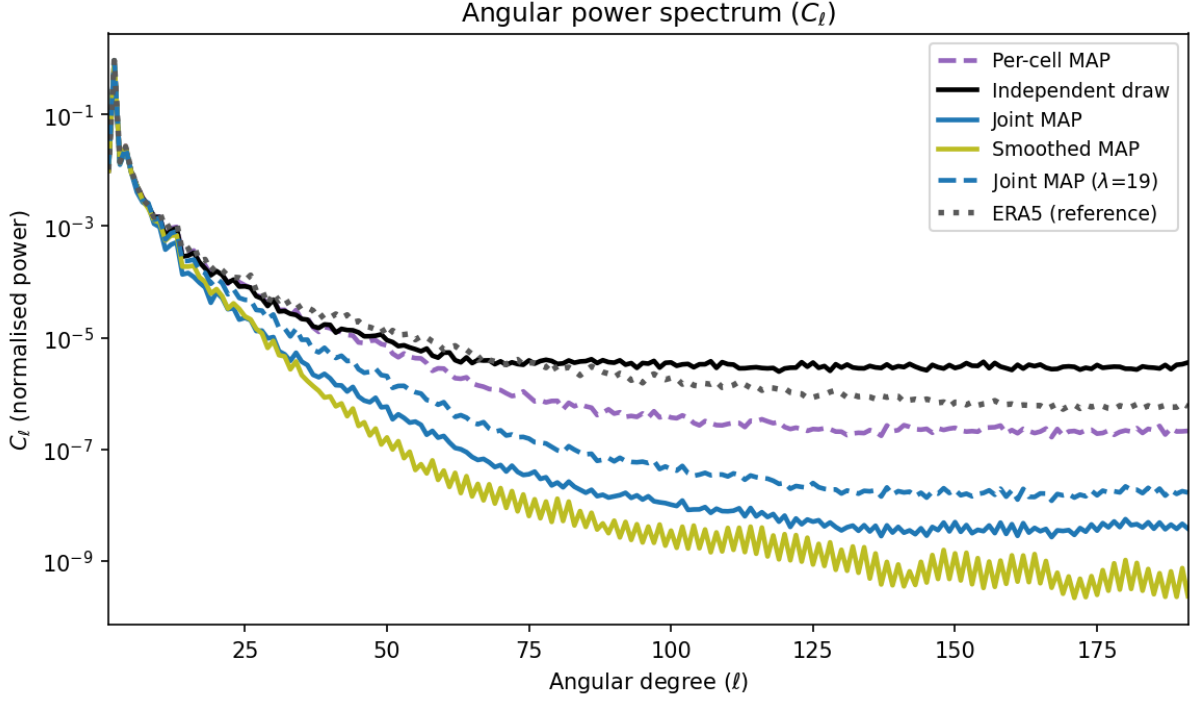


Figure 5.5: Angular power spectrum of the sampled +48 h fields (native O96 spherical-harmonic transform, 14-epoch model). C_ℓ against angular degree ℓ over the resolved band $\ell \in [1, \ell_{\text{res}}]$, monopole removed, with shapes compared rather than amplitudes as de-standardisation statistics are unavailable. The curves are the fields of Figure 5.2: the Independent draw is the over-noisy white floor at high ℓ ; the Per-cell MAP sits mid-band, below the grey ERA5 reference; Joint MAP at the budget setting $\lambda=19.0$ (dashed) lies between the Per-cell MAP and the matched pair; and the Smoothed MAP blur sits lowest, with Joint MAP at λ^* alongside it at matched $\tilde{R}$.

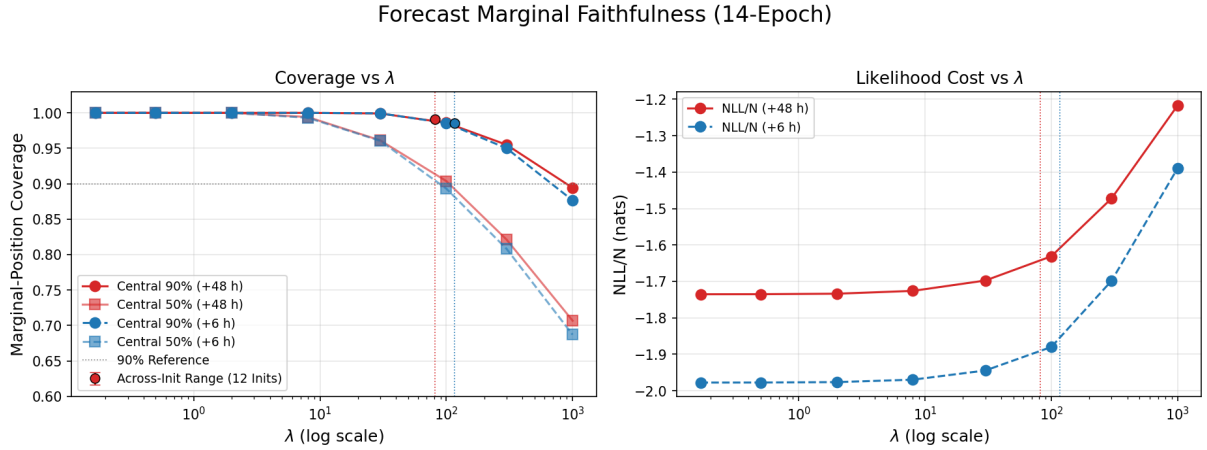


Figure 5.6: Joint MAP coverage along the λ sweep, 14-epoch model; the longest (+48 h, red) and shortest (+6 h, blue) steps. *Left*: central-90% and central-50% *marginal-position* coverage against λ (log axis), with λ^* (dotted) and the 0.90 reference level marked, the coverage an exact draw from each mixture attains in expectation. *Right*: the likelihood cost (NLL/ N under the emitted GMM) paid for coherence along the same sweep. Coverage stays near unity at λ^* whilst the likelihood cost rises only gently.

Chapter 6

Discussion and Conclusions

6.1 Sampler performance across synthetic and real settings

The synthetic testbed and the 48-hour forecast test the sampler in two regimes. The testbed’s mixture weights are often near-equal, whilst at +48 h one component still dominates most locations. The measured softening moves the forecasts toward the testbed’s regime as they reach further ahead.

Where weights are near-equal the most likely component changes from cell to cell, so the Per-cell MAP baseline is jagged and gives no usable field. Of the tested methods only Joint MAP reaches the smoothest settings (Figure 4.5). The price is that cells are pushed off their peaks, although this can be adjusted by varying the weight, and the loss stays fairly low here until the very smoothest settings.

Contrastingly, at +48 h the Per-cell MAP field is already consistent with its mixtures, and in Figure 5.2 its texture sits closest to ERA5’s, a comparison we read as direction-only, as we don’t reconstruct ERA5. Visually it slightly outperforms Joint MAP even at $\lambda = 19.0$, as temperature can change abruptly between neighbouring cells at this resolution. However, Joint MAP at λ^* still keeps more cells on their peaks than the Smoothed MAP blur in every initialisation, most clearly on the bimodal cells where a genuine choice exists. The smear tail of Figure 5.3 backs up this faithfulness.

In this report the matched-roughness target was a comparison device, not a recommended smoothness. In use, the weight could instead be tuned to the smoothness a given regime or resolution requires. Notably, the sampler’s smoothing naturally works against sharp weather structures, placing a limit on its ability.

6.2 Limitations

This research uses only one spatial resolution, one forecast variable and the specific WeatherGenerator GMM branch. The final model is also only trained for 14 forecast mini-epochs, without demonstrated convergence, so it is a more-trained checkpoint rather than a single settled trajectory (Appendix A). Notably, its validation likelihood and the share of well-separated bimodal cells were both still improving at the end of the final run, so the distributions studied are those of a model still visibly training. We test only out to a 48-hour forecast range, and the robustness check samples one start date per month of a single year, spanning that year’s four seasons but not several years or a longer climatology.

6.3 Future work

A strong point of this research is that the matched-roughness selection, the off-mode tail, the separate reading of the bimodal cells and the spectrum check apply to any future model on the same GMM branch, which itself had no evaluation architecture before this. The first next step would be to train the model to forecast seven to fourteen days out, where we expect the mixtures to further soften and spread. A preliminary seven-day training run showed steady weight softening, but no growth in well-separated multimodality. However, artefacts of longer forecasts are difficult to separate from undertraining, and it is likely a longer training run would help settle this. Joint MAP’s improvement over baselines should increase at these ranges, if the mixtures keep distinct modal points.

A much more work intensive step would be to sample several of the model’s output features simultaneously, asserting covariance between features such as rain and cloud cover, where ensemble copula coupling [13] and the Schaake shuffle [3] could become comparable baselines. A further direction would be to fine-tune the model to emit mixtures at finer target points, which should give more slowly varying mixtures, and to preserve sharp features like weather fronts.

6.4 Conclusions

This research produces a sampler that turns the independent per-location Gaussian mixtures a foundation weather model produces into a single field, true to each location’s own distribution whilst varying coherently across space. Validated on a synthetic testbed and tested on real 48-hour forecasts, it leaves far fewer values stranded off their emitted peaks than the Smoothed MAP baseline at matched smoothness.

Even at two days ahead, one component typically still carries roughly three-quarters of the mixture weight, so the improvement margin concentrates on the minority of cells

holding genuine alternatives. However, the mixtures soften as the forecast predicts further ahead, toward the fully softened regime seen in the synthetic testbed, the regime this sampler is built for, helping to produce forecasts that carry uncertainty through to the decisions they inform.

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Appendix A

Training configurations and hyperparameters

The Joint MAP minimisation of Section 3.3 uses four restarts of 400 steps at learning rate 0.05, retaining the lowest- J_λ field; each restart is jittered from the warm start by the median component spread of the loaded mixtures, with Adam’s default momentum parameters.

The two emitting models themselves were trained under the configurations of Figure A.1, the fourteen forecast mini-epochs spanning two warm-started runs, each mini-epoch a fixed-size shuffled subset of 4,096 training samples.

Shared training recipe (both regimes)

Trainable weights	whole model, nothing frozen
Loss	Gaussian-mixture negative log-likelihood on masked tokens
Mixture components	$K = 4$ weighted components per location
Training window	ERA5 1979–2022, 6-hour cadence
Validation window	October–December 2023
Optimiser	AdamW ($\beta_1 = 0.975$, $\beta_2 = 0.9875$, $\epsilon = 2 \times 10^{-8}$)
Weight decay / grad. clip	0.1 / 0.1
Learning rate	$10^{-6} \rightarrow 10^{-5}$; 512-step cosine warmup, 512-step linear cooldown

Per-regime settings

	Autoencoder	Forecast model
Initialisation	from scratch	warm start from the autoencoder checkpoint
Masking	random, rate 0.6	forecast (full current field as source)
Forecast steps	none (step-0 reconstruction)	eight 6-hour steps, to +48 hours
Mini-epochs	32 (checkpoint 31 used)	6 + 8 across two runs (run record in text)

Figure A.1: Training configurations for the two emitting models: the recipe shared by both regimes (top) and the per-regime settings (bottom). Each of the two forecast runs carried its own learning-rate warmup and cooldown.

Appendix B

Full regularisation and temperature sweeps

The main text’s operating points were selected from full sweeps over the smoothness weight λ and the mode-selection temperature β . The complete grids are the reconstruction-regime λ sweep on the real fields (Figure B.1) and the synthetic testbed’s full λ and β plane (Figure B.2), with the swept +48 h fields rendered in Figure B.3.

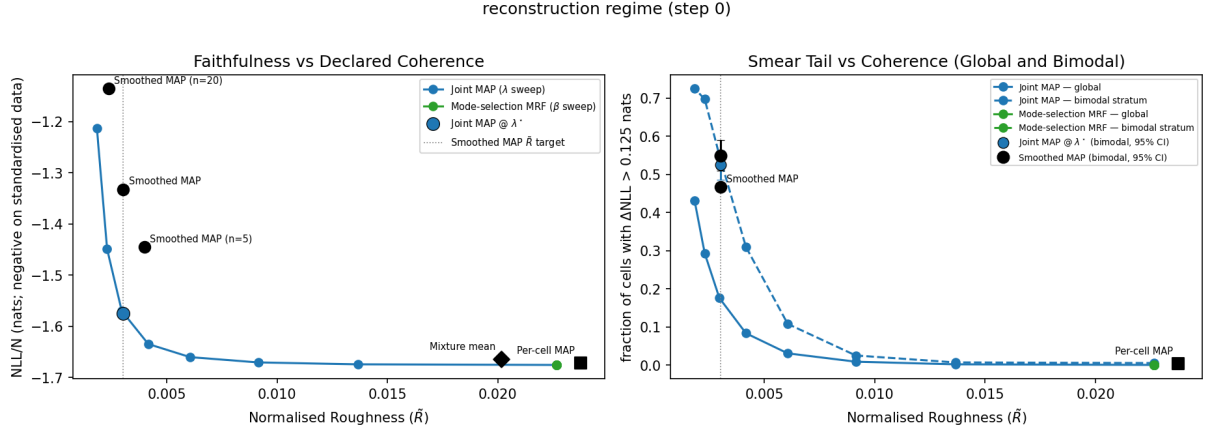


Figure B.1: As Figure 5.3, for the reconstruction-regime (step-0) full λ sweep; the Mode-selection MRF β sweep again collapses to a single point at the sweep origin in both panels.

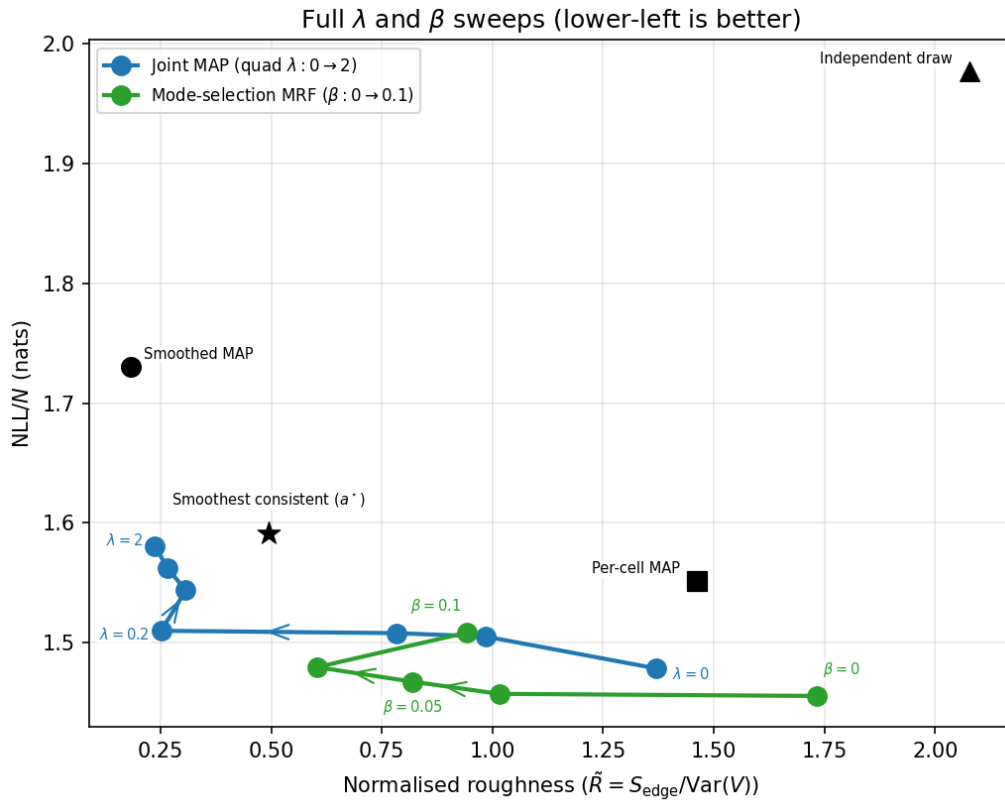


Figure B.2: The synthetic testbed's full sweep plane: every swept Joint MAP λ and Mode-selection MRF β configuration on the consistency-coherence plane, against the Independent draw, Per-cell MAP, Smoothed MAP and smoothest-consistent (a^*) anchors. The two penalty methods' operating points in Chapter 4 are selected from these grids.

Forecast +48 h: Joint MAP Lambda Sweep, North Atlantic / Europe

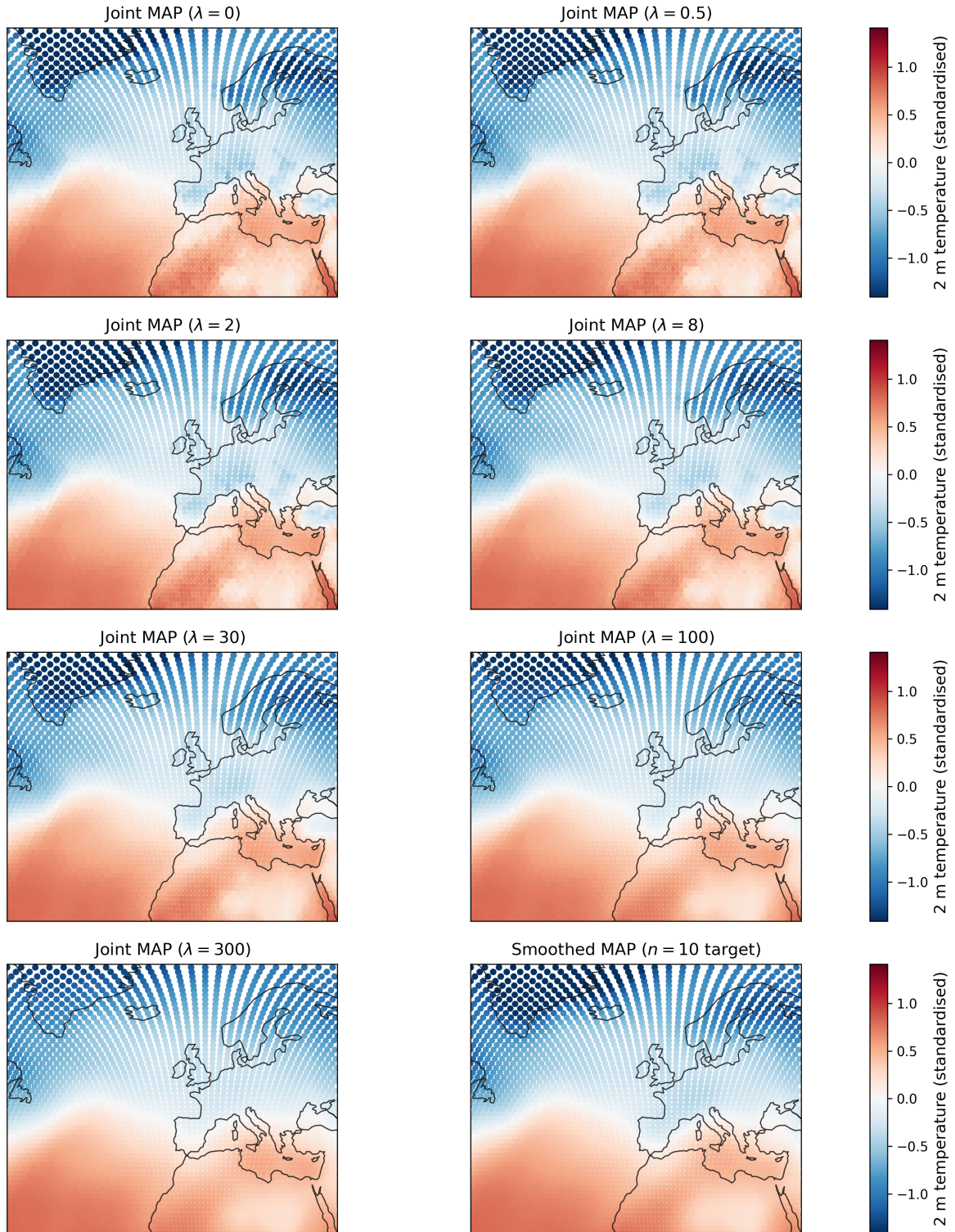


Figure B.3: Joint MAP across the regularisation sweep ($\lambda \in \{0, 0.5, 2, 8, 30, 100, 300\}$) for the +48 h snapshot of Figure 5.2, North Atlantic / Europe window, with the Smoothed MAP ($n=10$) roughness target alongside. The sweep starts at $\lambda=0$ from the refined per-location most-likely values and trades grid-scale texture for smoothness continuously as λ increases.

Appendix C

Additional ablations and example fields

Example-field maps extend the +48 h render of Chapter 5 to the reconstruction regime (Figure C.1) and the +6 h and +24 h steps (Figures C.2 and C.3), each at its step's own selected settings, with the whole-globe +48 h view in Figure C.4.

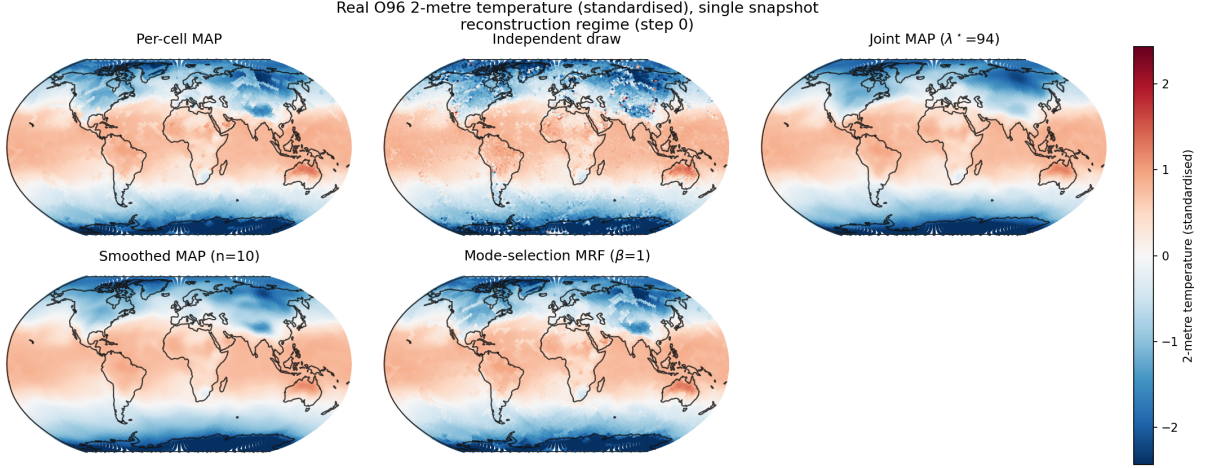


Figure C.1: Sampled fields for one unseen 2023 snapshot in the reconstruction regime (step 0, standardised units): the Per-cell MAP and Independent draw anchors, Joint MAP at this step's λ^* with the Smoothed MAP blur, and Mode-selection MRF beside ERA5. With the mixtures nearly single-valued at step 0, the five fields differ visibly only in the Independent draw speckle.

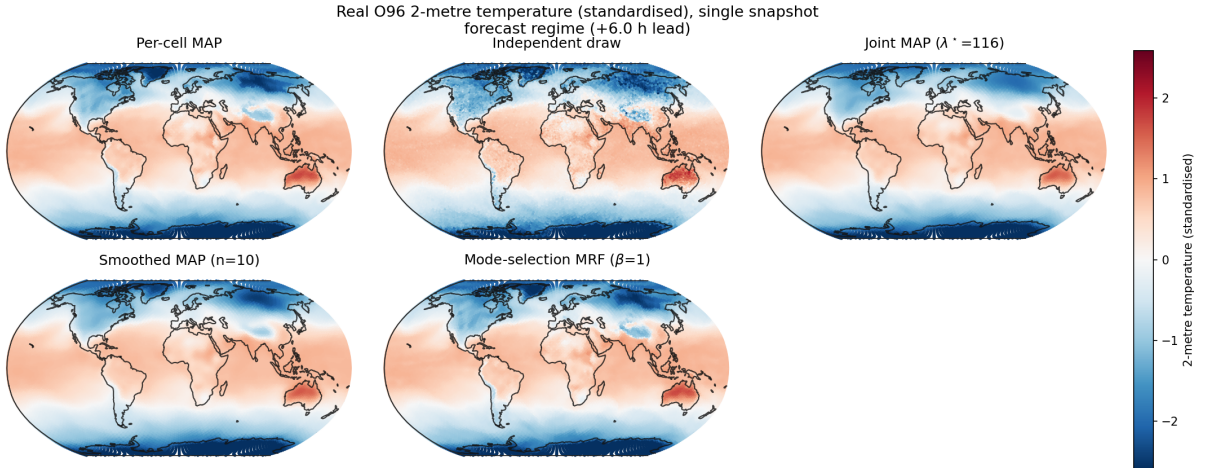


Figure C.2: Sampled +6 h fields for the same snapshot (14-epoch model, standardised units), in the panel order of Figure C.1. The Joint MAP panel sits at this step's matched λ^* (Table 5.2); the panel headings carry the selected values.

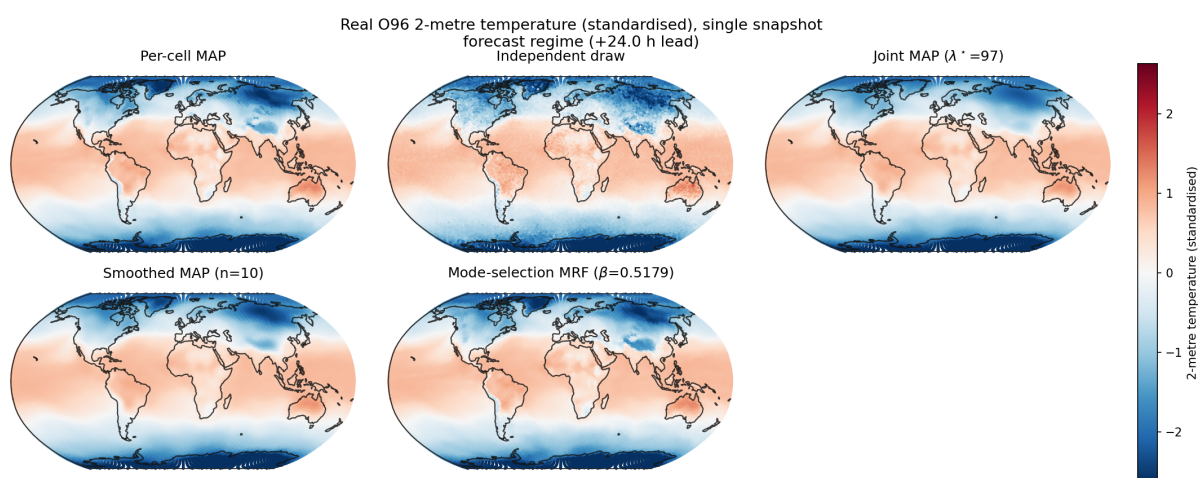


Figure C.3: As Figure C.2, at the +24 h step: the same five methods at this step's matched λ^* (Table 5.2) and its Mode-selection MRF temperature, with the selected values in the panel headings.

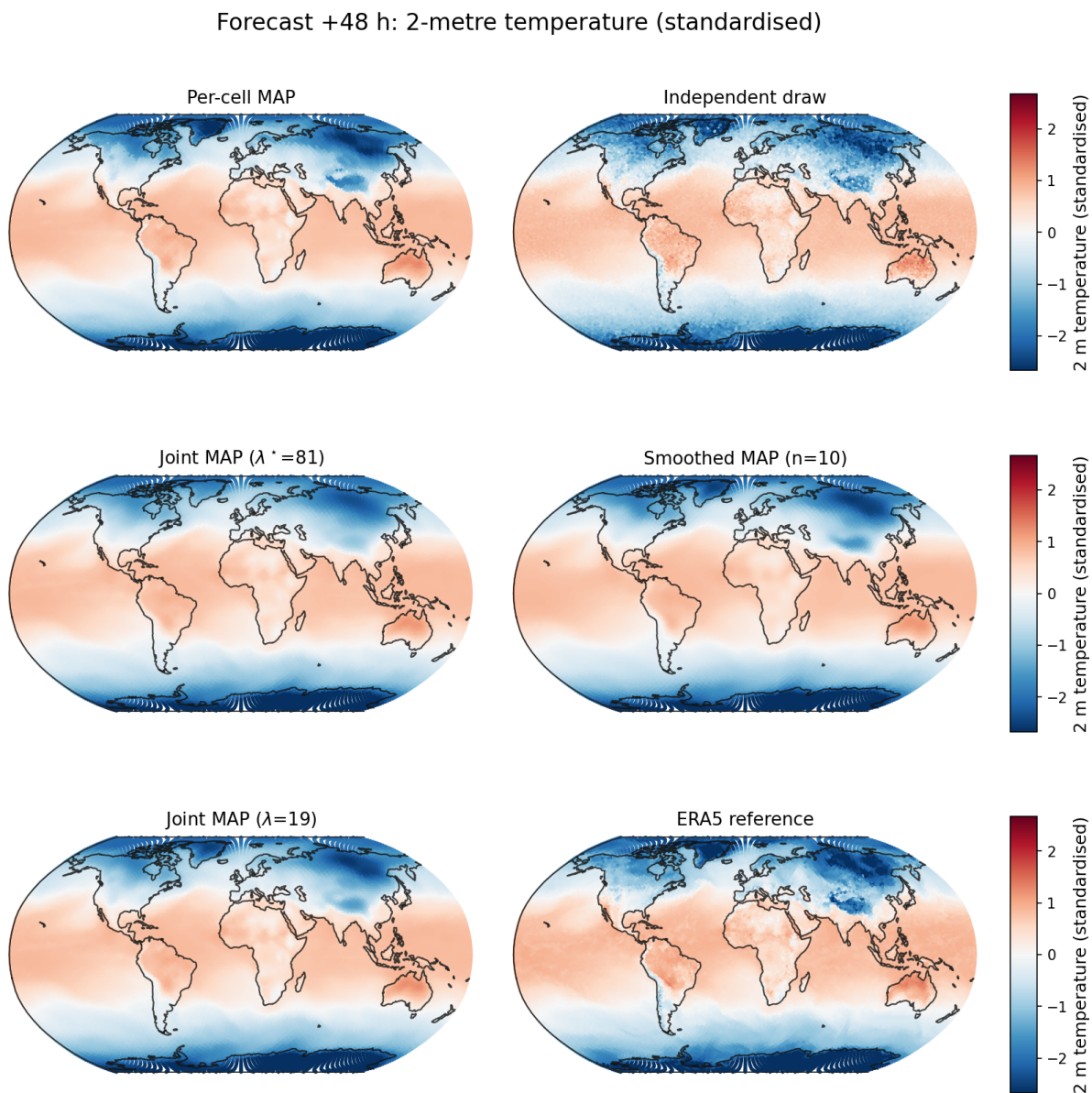


Figure C.4: Whole-globe (Mollweide) view of the sampled +48 h fields of Figure 5.2, same panels and standardised units. Grid-scale texture is compressed at this projection, which is why the main text shows the regional window.

Appendix D

Numerical implementation notes

Mixture parameters are stored in single precision, so the Gaussian density underflows to zero beyond about 13σ from a component mean,

$$z_{\max} = \sqrt{2 \ln(1/\text{FLT_MIN})} \approx 13.2,$$

and likelihood evaluation runs in log space via `logsumexp`. A dense graph Laplacian over all $N = 40,320$ locations would need about 13 GB, so the smoothness penalty runs on a sparse k -nearest-neighbour graph ($k=8$), symmetrised by the union of neighbourhoods.

The Joint MAP descent of Section 3.3 uses the consistency term’s closed-form gradient,

$$\frac{\partial \text{NLL}}{\partial x_i} = - \sum_{k=1}^K r_{ik} \frac{\mu_{ik} - x_i}{\sigma_{ik}^2}, \quad r_{ik} = \frac{\pi_{ik} \mathcal{N}(x_i; \mu_{ik}, \sigma_{ik})}{p_i(x_i)}, \quad (\text{D.1})$$

where x_i is the value chosen at location i .

Two optimality certificates from the synthetic testbed back Adam’s solutions as the Joint MAP operating points. An exact min-cut solve of the discretised quadratic objective [8] matches Adam exactly at every $\lambda \geq 0.5$ and sits within the restart floor over $\lambda = 0.1$ – 0.2 . The certified quadratic optimum still smears whilst the exact total-variation solution collapses instead, its gradient optimiser failing the certificate at every $\lambda \leq 0.2$, so smearing is the objective’s property rather than the optimiser’s.

Appendix E

Reproducibility and compute

Each training run used a single node of the Alps Clariden HPC system (CSCS), 11.8 wall-clock hours for the autoencoder and 10.1 for the forecast run; extraction from a checkpoint takes about 1.2 minutes. The full real-data audit, both method sweeps plus the graph build, completes in about 77 seconds on one CPU. Results are deterministic given the emitted artifacts, from fixed seeds throughout.

The sampler and analysis code lives in a dedicated private repository, accessible to the examiner and supervisor, at <https://gitlab.developers.cam.ac.uk/phy/data-intensive-science-mphil/assessments/projects/hb747>; the model itself is the WeatherGenerator codebase [14]. A step-by-step manual for reproducing the emitting models, from data access to the training and extraction commands, is provided in the repository (`docs/reproducing_the_model.md`).

Appendix F

Declaration of use of autogeneration tools

This dissertation was prepared with the assistance of AI-based autogeneration tools (large language model assistants). All research design, modelling decisions, analysis, numerical results and scientific claims are my own.

The tools were used for the following purposes.

Software and figures

- Assisting in drafting and reviewing unit and integration tests for the sampler and analysis code.
- Advising on the structure and organisation of the code repository.
- Generating standard Python docstrings for functions and classes.
- Producing boilerplate visualisation code (Matplotlib/Seaborn scaffolding for axes, legends and colour maps, in particular for the spatial-field figures).
- Assisting in the production of the report's plots and figures from my own results.
- Writing boilerplate shell scripts for HPC job submission, for example the warm-start scripts used to continue forecast training (`continue_gmm_fc48.sh`).
- Suggesting more idiomatic (Pythonic) rewrites of existing code and fixing linting errors.

Report

- Formalising the report language, checking for mistakes and general errors across the chapters, as well as help structuring and organising the report.
- Debugging LaTeX including compilation errors and building complex tables.